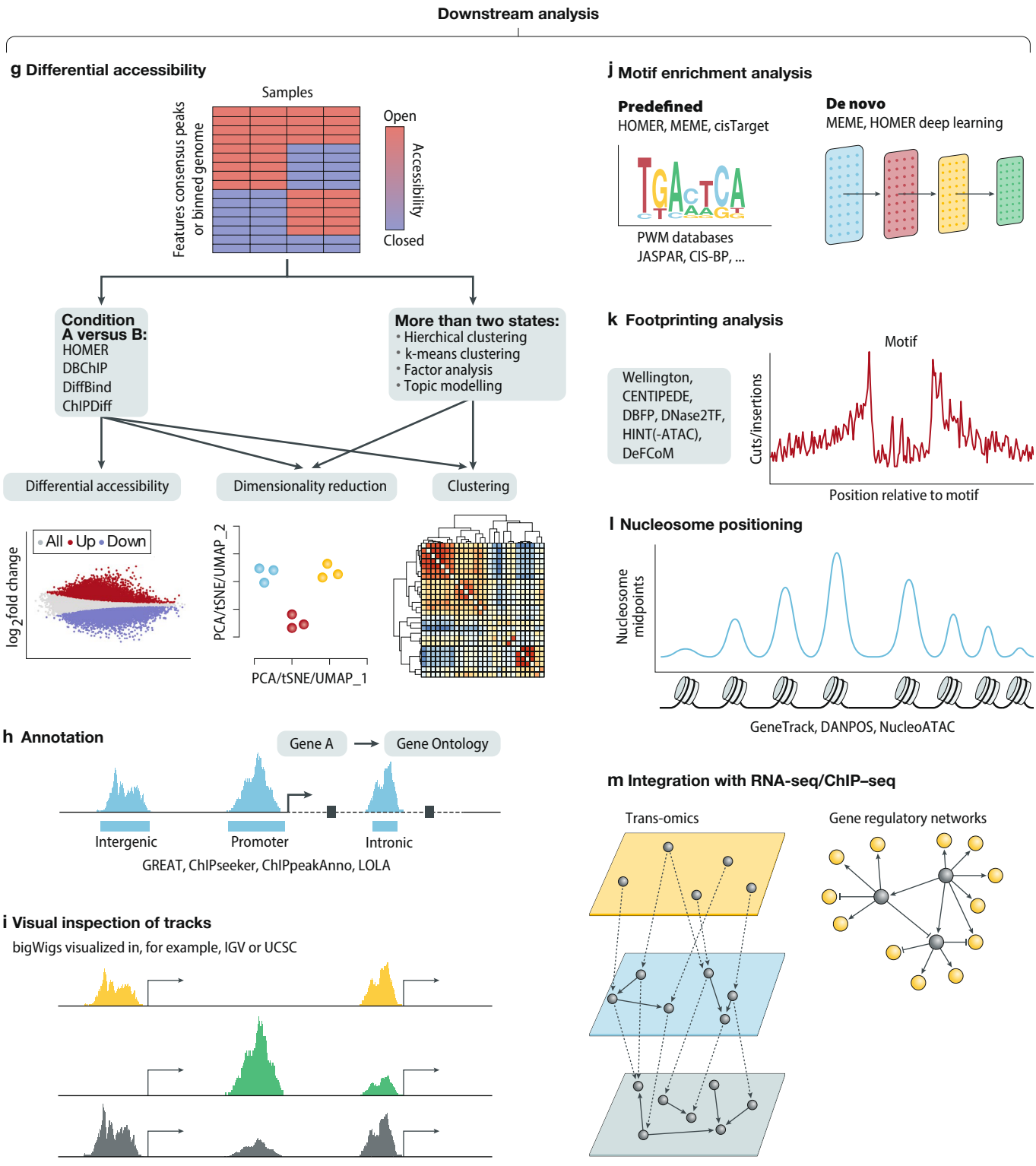

Downstream analysis
g Differential accessibility
Samples
Features consensus peaks or binned genome
Open
Accessibility
Closed
Condition A versus B:
HOMER
DBChIP
DiffBind
ChIPDiff
More than two states:
• Hierchical clustering
• k-means clustering
• Factor analysis
• Topic modelling
Differential accessibility
Dimensionality reduction
Clustering
All • Up • Down
log2fold change
PCA/tSNE/UMAP_2
PCA/tSNE/UMAP_1
h Annotation
Gene A
Gene Ontology
Intergenic
Promoter
Intronic
GREAT, ChIPseeker, ChIPpeakAnno, LOLA
i Visual inspection of tracks
bigWigs visualized in, for example, IGV or UCSC
j Motif enrichment analysis
Predefined
HOMER, MEME, cisTarget
PWM databases
JASPAR, CIS-BP, ...
De novo
MEME, HOMER deep learning
k Footprinting analysis
Wellington, CENTIPEDE, DBFP, DNase2TF, HINT(-ATAC), DeFCoM
Motif
Cuts/insertions
Position relative to motif
l Nucleosome positioning
Nucleosome midpoints
GeneTrack, DANPOS, NucleoATAC
m Integration with RNA-seq/ChIP–seq
Trans-omics
Gene regulatory networks